



Lesson 2.10 — Unit 2 Review and Assessment

Big idea: Unit 2 is the three forms of a quadratic, three ways to solve them (factor, complete the square, formula), and the discriminant that tells you what kind of solutions to expect.

Quick review

Three forms: standard ax^2+bx+c (gives y-intercept), vertex $a(x-h)^2+k$ (gives vertex), factored $a(x-r_1)(x-r_2)$ (gives zeros).

Three solving methods: factor (when easy), complete the square (always works, builds vertex form), quadratic formula (always works, gives exact form).

Discriminant $b^2 - 4ac$: $> 0 \rightarrow$ two real, $= 0 \rightarrow$ one repeated, $< 0 \rightarrow$ two complex.

Complex numbers: $i^2 = -1$; conjugates kill the imaginary part when multiplied.

Worked example

Worked review. $y = 2x^2 - 8x + 5$.

Vertex form: $2(x-2)^2 - 3$, vertex $(2, -3)$. Standard zeros via formula: $x = (8 \pm \sqrt{24})/4 = 2 \pm \sqrt{6}/2$.

Warm-ups

1. State the vertex of $y = (x-4)^2 + 1$.
2. Solve $x^2 - 5x + 6 = 0$ by factoring.
3. Compute the discriminant of $3x^2 - 4x + 2 = 0$.

Practice

1. Convert $y = x^2 + 6x + 11$ to vertex form.
2. Solve $2x^2 + 3x - 5 = 0$.
3. Simplify $(3 - 2i)(1 + 4i)$.

Challenge

1. Find the value of c so that $x^2 - 8x + c = 0$ has a double root.
2. A toy rocket's height is $h(t) = -16t^2 + 64t$. Find max height, time it occurs, and time of landing.



Answer key — Lesson 2.10

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $(4, 1)$.
2. $(x - 2)(x - 3) = 0 \Rightarrow x = 2, 3$.
3. $16 - 24 = -8$ (negative $\rightarrow$ complex roots).

Practice

1. $y = (x + 3)^2 + 2$.
2. Factor $(2x + 5)(x - 1) = 0 \Rightarrow x = -5/2, 1$.
3. $3 + 12i - 2i - 8i^2 = 3 + 10i + 8 = 11 + 10i$.

Challenge

1. $D = 64 - 4c = 0 \Rightarrow c = 16$.
2. Vertex $t = 64/32 = 2$ s, max $h = -64 + 128 = 64$ ft. Lands when $-16t(t - 4) = 0 \Rightarrow t = 4$ s.